

ALIBABA CLOUD AI HACKATHON 2026

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ROZGAR AI

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Voice-First Job Marketplace for Pakistan's Informal Workforce

PROJECT REPORT

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Live Demo: <https://rozgar-ai.streamlit.app/>

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Abstract

Rozgar AI is a fully voice-driven job-matching platform designed for Pakistan's informal workforce — electricians, tailors, domestic helpers, drivers, security guards, and millions of similarly skilled workers who are excluded from mainstream digital job platforms because those platforms assume typing ability, English literacy, and a formal CV. Rozgar AI removes every one of these barriers: a worker or an employer simply speaks, in Urdu, and the system builds a structured profile or job listing automatically.

The system combines OpenAI Whisper for Urdu speech-to-text transcription, Google Gemini for intelligent field extraction (with a fully offline rule-based fallback for resilience), a multilingual sentence-transformer model for semantic skill matching, and a custom weighted scoring engine that adapts its priorities — location, skill, or experience — based on job category. The entire interface is rendered in pure Urdu script, and the system is built to run at effectively zero infrastructure cost, making it realistic to deploy at scale in low-resource environments.

This report documents the problem being solved, the system architecture, the matching methodology, the current feature set with live dashboard evidence, the intended real-world impact, and the roadmap toward a full production release.

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1. Introduction

1.1 The Problem

Pakistan's informal economy employs tens of millions of skilled workers — tradesmen, home-based women workers, and bulk-staffing labour — yet almost every digital job platform available today is built for a white-collar, English-typing, CV-holding user. A tailor in Korangi or a security guard in Malir does not need a resume builder or a job-application form; she or he needs to speak once, in their own language, and be discovered by the right employer nearby.

This mismatch keeps a large, economically active population locked out of the efficiencies that digital platforms otherwise bring to job matching — faster discovery, wider reach, and fairer price transparency.

1.2 The Solution

Rozgar AI closes this gap by making voice the only required input. A worker or employer records a short description of themselves or their need; the system transcribes it, extracts structured fields such as skill, location, experience, and rate using AI, and immediately matches it against the opposite side of the marketplace using a transparent, weighted scoring algorithm — all without requiring a single typed word.

1.3 Objectives

- Eliminate literacy and English-language barriers to digital job discovery.
- Provide a structured, searchable profile from a single voice recording.
- Match workers and employers using a scoring method that reflects real-world priorities per job category.
- Operate reliably even without consistent internet access, via an offline fallback pipeline.
- Keep the entire stack low-cost enough to be sustainable for a nonprofit or community deployment.

2. System Architecture

Rozgar AI follows a linear, modular pipeline. Each stage has a single responsibility, which keeps the system easy to debug, extend, and — critically — keeps it functional even when a component such as the Gemini API is temporarily unavailable.

Stage	Function
1. Voice Input	Worker or employer records audio, or uploads an existing clip (MP3 / WAV / M4A / OGG / MP4).
2. speech_to_text.py	OpenAI Whisper ('small' model) transcribes the Urdu audio into text, with broad exception handling for missing codecs.
3. llm_extraction.py	Google Gemini extracts structured fields — skill, location, experience, rate, working hours, description — from free-form speech.
4. extraction.py (fallback)	A rule-based, keyword-matching extractor takes over automatically if Gemini is unreachable, so the app never fully breaks offline.
5. storage.py	Structured entries are persisted to JSON storage (workers.json / employers.json) — a lightweight, dependency-free data layer.
6. matching.py	A weighted scoring engine ranks the opposite side of the marketplace by location, skill, and experience relevance.
7. app.py	A pure Urdu-script Streamlit interface presents results, editable profile cards, and category browsing.

A key design decision is the automatic fallback from Gemini to the rule-based extractor: if the AI service is unreachable due to no internet, quota limits, or a model-name change, the rule-based pipeline seamlessly takes over so a worker's profile is never lost, and the failure is invisible to the end user.

3. Technology Stack

Layer	Technology
Speech-to-Text	OpenAI Whisper — 'small' model, tuned for Urdu
Smart Extraction	Google Gemini API, with multi-model-name fallback and caching
Offline Fallback	Custom rule-based keyword/clause extraction engine (pure Python)
Semantic Matching	paraphrase-multilingual-MiniLM-L12-v2 sentence-transformer
Data Storage	JSON file storage (workers.json / employers.json)

Interface	Streamlit — fully Urdu-script UI
Voice Playback	gTTS text-to-speech for 'Suniye' listing playback
Offline Sharing	QR code generation for printable flyers
Hosting	Streamlit Community Cloud — zero infrastructure cost

4. Matching Methodology

Rather than a single generic ranking formula, Rozgar AI scores every match using a category-aware weighted average, reflecting the fact that different trades have fundamentally different priorities.

4.1 Scoring Formula

Final Score = (Location Score × W_location) + (Skill Score × W_skill) + (Experience Score × W_experience)

4.2 Category Weight Profiles

Category	Location	Skill	Experience
Tradesman	0.3	0.5	0.2
Home-Based (Women)	0.6	0.4	0.0
Bulk Staffing	0.2	0.5	0.3

Home-based women workers (e.g. tailors) are weighted heavily toward location, because their customers travel to them. Tradesmen are weighted toward skill, since they travel to the job. Bulk-staffing roles favour skill and experience, reflecting urgency and worker mobility.

4.3 Location Proximity

- Same area → 1.0
- Same district → 0.6
- Different district → 0.3
- Either area unknown → 0.3 (neutral — does not penalise missing data)

Karachi's 7 official districts are used as a distance proxy, avoiding any dependency on GPS or paid maps APIs.

4.4 Skill Scoring

Skill matching is tiered rather than relying on raw embedding similarity alone, which would let unrelated trades score deceptively high purely from similar sentence phrasing. An exact skill match scores 1.0; a different skill within the same category is capped at 0.5; a different category entirely is capped at 0.25 — all computed via a multilingual sentence-transformer so that differently worded Urdu or Roman-Urdu descriptions of the same skill still score highly against each other.

4.5 Honest, Non-Overconfident Results

Any match scoring below 0.4 is explicitly flagged as a weak match and displayed to the user as 'closest available, not an exact match' rather than being presented with false confidence. The engine never returns zero results if any opposite-side entries exist — it always shows the best available option, honestly ranked.

5. Key Features

5.1 Implemented and Tested

- Voice recording and file upload, with playback confirmation before submission
- Urdu speech transcription via Whisper
- Dual extraction: Gemini (smart) with automatic offline rule-based fallback
- Multi-model fallback for Gemini — tries candidate model names and caches the working one
- Editable profile cards — every field individually correctable without a full re-record
- Weighted, category-aware matching engine
- Phone number masking (last 4 digits visible, reveal-on-click) for privacy
- Pure Urdu-script interface — every label, button, and tab
- QR code generation and WhatsApp share for printable, offline-reachable flyers
- Text-to-speech 'Suniye' playback of listings

5.2 In Progress / Schema-Ready

- Free-text rate and working-hours capture (never forced into a misleading single number)
- Applicant counter and 'urgent' job flag (backend complete, UI polish ongoing)
- Relative 'posted X ago' timestamp display

6. Application Walkthrough

The following screenshots are captured from the live, deployed build of Rozgar AI, demonstrating the end-to-end voice-first flow, from onboarding through to a ranked match.

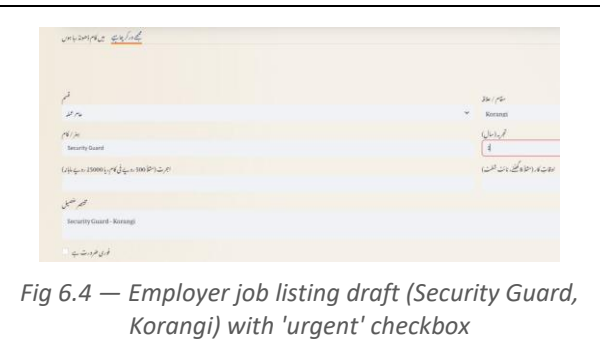
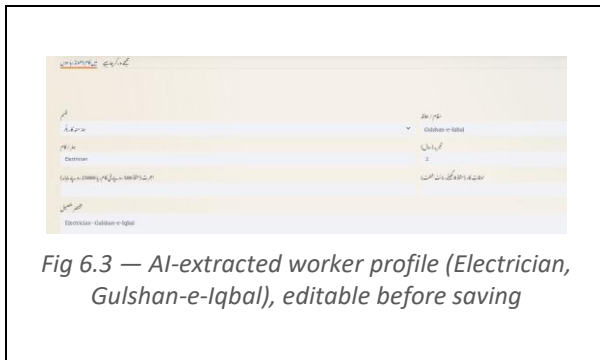
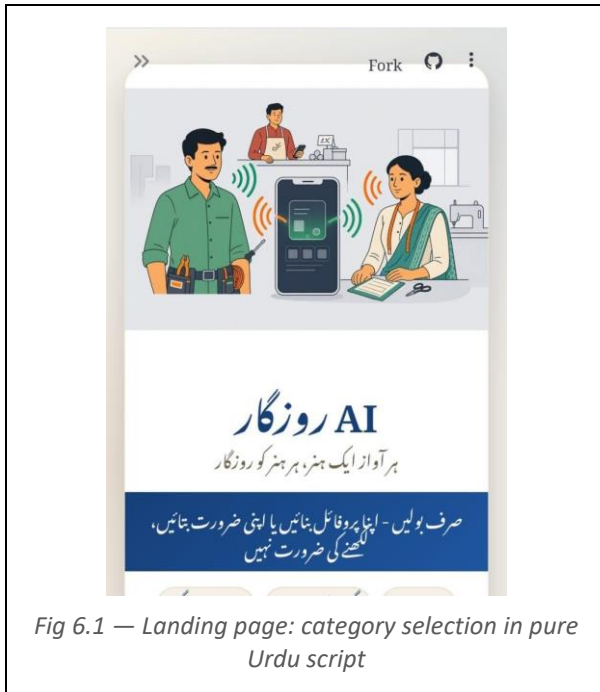




Fig 6.5 — Ranked matches with transparent match-percentage scoring



Fig 6.6 — Browsing tradesman listings by category



Fig 6.7 — Live marketplace statistics and filters



Fig 6.8 — QR code and WhatsApp share for offline reach

7. Impact and Beneficiaries

7.1 Who Benefits

- Skilled tradesmen (electricians, plumbers, AC technicians, mechanics, carpenters, painters, welders) who can now be discovered without writing a single word.
- Home-based women workers (tailors, home tutors, embroiderers, cooks) who gain visibility to nearby customers while working from home.

- Bulk-staffing workers (security guards, labourers, domestic helpers, drivers, cleaners, delivery staff) who often face the highest literacy and access barriers of all.
- Employers and households, who gain a faster, more transparent way to find verified, locally available help.
- The broader informal economy, where better matching reduces underemployment and wage opacity.

7.2 Why This Matters

Pakistan's informal labour market is enormous but chronically inefficient — workers and employers rely on word-of-mouth, physical labour chowks, or informal contacts, which limits both reach and fairness. By removing the literacy and typing barrier entirely, Rozgar AI extends the reach of digital job discovery to a population that mainstream platforms structurally exclude, directly supporting the Alibaba Cloud AI Hackathon's mission — delivered in partnership with Bano Qabil and the Alkhidmat Foundation — of practical, accessible AI for underserved communities.

8. Results

The deployed demo build, seeded with realistic Karachi-based data across four districts (Karachi East, Karachi Central, Korangi, and Malir), demonstrates the full pipeline end-to-end.

Metric	Demo Value
Registered workers	61
Available jobs	27
Available workers	59
Joined today (session)	5
Top match example	Security Guard, Gulshan-e-Iqbal — 71% match

The matching engine consistently surfaces the most relevant candidate first, weighting exact skill and area matches highest, while still returning ranked, honestly-labelled results even in sparse-data areas.

9. Future Scope

- **Voice-Based Availability Toggle** — Currently, marking oneself as "Available/Busy" requires a single button tap; in a future version, this could also be set via a voice command (using a lightweight classification step), so that a worker never has to type or click at all.
- **Area-Based Stats Page** — A dedicated analytics page showing insights such as "how many jobs are available in Lyari" or "citywide demand for tutors across Karachi." This was kept out of the current scope as it represents an entirely new feature area.
- **Alibaba Cloud Migration** — The current build uses OpenAI Whisper and Google Gemini; the production version could migrate to Alibaba Cloud's native services — Paraformer for speech-to-text, Qwen/DashScope for field extraction, and ApsaraDB/Tablestore for storage — aligning fully with the hackathon's sponsor ecosystem.
- **Multi-City Expansion** — Currently limited to 4 districts of Karachi; future versions could expand to cover all of Karachi, then other cities such as Lahore and Islamabad, scaling with growing data density and user base.
- **Trust Score** — Derived from applicant interest count, to help employers gauge a worker's credibility at a glance.

Explicitly Out of Scope (by design)

- In-app chat/messaging — Call and WhatsApp buttons cover the same need with far less complexity
- A full LinkedIn-style profile page — this would contradict the platform's core 'no CV, no form' philosophy

10. Conclusion

Rozgar AI demonstrates that a genuinely voice-first, zero-typing job marketplace is technically achievable today, using a combination of speech recognition, LLM-based extraction, semantic matching, and a transparent scoring engine — all deployable at effectively zero infrastructure cost. By designing for the specific realities of Pakistan's informal workforce — low literacy, no English requirement, unreliable connectivity, and category-specific priorities — the system aims to make digital job discovery genuinely accessible rather than merely translated.

The current build already delivers a complete, working voice-to-match pipeline with an honest, explainable scoring system, and a clear, prioritised roadmap toward the remaining production features. It was built and submitted for the Alibaba Cloud AI Hackathon 2026, in partnership with Bano Qabil and the Alkhidmat Foundation.

Acknowledgement

Developed by Abrar Shakeel Ghoury for the Alibaba Cloud AI Hackathon 2026.